

KINJAL SINGH

Computer Science & Statistics at UIUC

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EDUCATION: University of Illinois at Urbana-Champaign (UIUC)

Expected May 2026

Bachelor of Science in Computer Science and Statistics

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (Node.js, React), SQL, Rust, Bash

Full-Stack & Web: React, Node.js, Express, Flask, REST APIs, gRPC, HTML, CSS, TypeScript

Backend & Distributed Systems: Kafka, Redis, MongoDB, Hadoop, ELK Stack

Database Systems: PostgreSQL, DynamoDB, Redshift, Snowflake, BigQuery, Athena, Apache Spark, PySpark, Airflow, AWS Glue

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Helm, Terraform, GitHub Actions (CI/CD), Linux, Istio

Machine Learning: Scikit-learn, PyTorch, TensorFlow, XGBoost, MLflow, Hugging Face

WORK EXPERIENCE

ADAS Validation Intern - Rivian

September 2025 – Present

- Collaborating with the **SoC validation team** to execute validation plans for next-gen ADAS/embedded hardware platforms.
- Developing and running **C validation tests** for SoC boot sequences and **peripheral interfaces (UART, I2C, SPI, Ethernet, DDR, PCIe)**, improving functional coverage by **28%**.
- Designing **benchmark tools** using **Python** to evaluate throughput, latency, resource utilization and I/O frequency, identifying bottlenecks that boosted validation efficiency by **30%**.

Software Engineer Intern - PricewaterhouseCoopers (PwC)

July 2025 – August 2025

- Designed a self-healing **Agentic AI Ops platform** to detect & remediate cloud/on-prem infra incidents on data centers via real-time log pipelines (**Prometheus, ELK, Kafka**), streaming **100K+ log events/sec**.
- Implemented **C++ gRPC microservices** for LLM inference, optimizing **ONNX Runtime** and cutting latency **4×** compared to Python-only deployments.
- Deployed **predictive scaling** using **ARIMA & Prophet** integrated with **Kubernetes HPA** and **Terraform**, reducing downtime by **35%** and manual remediation by **70%**.

NLP Research Assistant- Respeaker Project at UIUC

February 2024 – Present

- Built an **edge software pipeline** with **Raspberry Pi** and **ReSpeaker** to process **40+** hrs of classroom audio, enabling real-time analytics of classroom discussions through speaker diarization and engagement tracking.
- Engineered **Python** services with **Flask APIs** for diarization and sentiment tracking, integrating fine-tuned **DistilBERT & Way2Vec2**, and achieved a model performance increase from **65% → 78% F1 score**.
- Optimized preprocessing and batch pipelines with **Python, Spark**, and **Airflow**, cutting end-to-end latency by **35%**, while scaling backend queries to **<200ms** on **BigQuery** and **Athena** to support interactive classroom analytics.

Data Engineering Intern - Clover Infotech

May 2024 – July 2024

- Engineered distributed **ETL workflows** with **PySpark, Hadoop**, and **AWS Glue**, consolidating **5+ enterprise data sources** into **Snowflake and Redshift** for real-time analytics.
- Developed **JavaScript-based Lambda functions** for dynamic schema evolution & data validation, reducing ETL failures by **25%**.
- Built a **Python** and **React** service exposing ETL metadata as **REST APIs** for BI dashboards, enabling **~1M daily API requests**.
- Improved pipeline observability with **Prometheus** and **Grafana** and **Linux-based log rotation strategies**, reducing incident response time by **40%**.

PROJECT HIGHLIGHTS

AutoLitigator – Agentic AI Legal Document RAG API Platform

January 2025 - March 2025

- Led a **3-member team** to design and deploy a **retrieval-augmented(RAG) backend (Pinecone and Redis)** handling **10K+** documents with **sub-500ms query latency**.
- Built scalable microservices (**LangChain** and **FastAPI**) containerized on **Kubernetes (EKS)** with **HPA autoscaling**, sustaining **1K+** concurrent requests at **95% uptime**.
- Exposed core services via **REST APIs** and built a **React** and **Node.js frontend** for interactive legal case search and analytics.

CERTIFICATIONS: AWS: Machine Learning Specialty, Data Engineer Associate, AI Practitioner, Cloud Practitioner

Google Cloud: Professional Data Engineer, Cloud Gen AI Leader, Digital Leader

